Master Glossary

All terms defined across all chapters, listed alphabetically with their source chapter.

Term	Definition	Chapter

8

80/20 rule (Pareto principle)	~80% of outcomes come from ~20% of effort or causes	Ch 10

A

Ablation study	Systematic removal of model components to measure their individual contributions	Ch 10
ACF	Autocorrelation function — autocorrelation plotted at every lag	Ch 02
Activation function	A nonlinear function (ReLU, sigmoid) applied to neuron outputs; enables nonlinear learning	Ch 07
Adaptive conformal prediction	A variant that scales corrections by the model's predicted uncertainty	Ch 08
AEMO	Australian Energy Market Operator — runs the NEM dispatch and settlement	Ch 01
Air mass	Ratio of atmospheric path length to the minimum path	Ch 04
arcsinh	Inverse hyperbolic sine — the target transform for electricity prices	Ch 01
Asymmetric calibration	Targeting different coverage levels for upper and lower tails	Ch 08
Autocorrelation	Correlation of a time series with its own past values	Ch 02

Autoregressive feature	Feature constructed from past values of the target itself	Ch 05

B

Backpropagation	The algorithm for computing gradients and updating weights in neural networks	Ch 07
Batch normalisation	A technique that normalises layer inputs to stabilise and accelerate training	Ch 07
Battery cycle	One complete charge-discharge (or equivalent energy throughput)	Ch 09
Battery degradation	Gradual loss of capacity due to cycling, aging, and operating conditions	Ch 09
Battery duration	Hours the battery can sustain maximum output (energy/power ratio)	Ch 09
Betz limit	Theoretical maximum wind energy extraction efficiency (~59.3%)	Ch 04
Bias	Error from overly simplistic model assumptions	Ch 06
Bias-variance tradeoff	The tension between model simplicity (high bias) and sensitivity to data (high variance)	Ch 07
Black-box model	A data-driven model that learns relationships without incorporating domain knowledge	Ch 10
Blade pitch control	Rotating blades to regulate power capture	Ch 04
Breakeven spread	The minimum price ratio between discharge and charge for profitable operation	Ch 09

C

Calibration (reliability)	Whether stated probabilities match observed frequencies	Ch 08
Calibration window	The period of recent data used to compute conformal corrections	Ch 08
Calibration-sharpness paradigm	"Maximise sharpness subject to calibration" — the guiding principle	Ch 08
Capacity factor	Ratio of actual output to theoretical maximum	Ch 04
Capacity payments	Revenue for being available to generate, regardless of actual dispatch	Ch 09
Capture ratio	Forecast revenue / perfect foresight revenue	Ch 05
Capture ratio (CR)	Actual dispatch revenue divided by perfect-foresight revenue	Ch 09
CCGT	Combined Cycle Gas Turbine — efficient gas plant	Ch 03
CDF (cumulative distribution function)	A function giving the probability of being at or below each value	Ch 08
Chance constraint	A constraint required to hold with at least a specified probability	Ch 09
Clear-sky model	Model predicting irradiance under cloud-free conditions	Ch 04
Closed-loop control	Adjusting plan as new information arrives	Ch 05
Cloud enhancement	Irradiance exceeding clear-sky due to cloud reflections	Ch 04
Coefficient	Weight on a feature in a linear model	Ch 06

Concavity	Diminishing marginal returns — each additional unit of input yields less additional output	Ch 09
Concept drift	Change in feature–target relationships over time	Ch 06
Conformal prediction	A model-agnostic framework for constructing prediction intervals with coverage guarantees	Ch 08
Conformity score	A measure of how poorly a model's prediction matches the actual outcome	Ch 08
Copula	A function describing the dependence structure between random variables	Ch 09
Cross-validation	Estimating generalisation error by training/validating on data splits	Ch 06
CRPS	Continuous Ranked Probability Score — the standard metric for probabilistic forecasts	Ch 08
CSI	Clear-Sky Index — ratio of actual to clear-sky irradiance	Ch 04
Cut-in speed	Minimum wind speed for turbine generation (~3 m/s)	Ch 04
Cut-out speed	Maximum safe operating wind speed (~25 m/s)	Ch 04
cvxpy	Python library for convex optimisation	Ch 05

D

Data assimilation	Mathematical fusion of model predictions and real observations	Ch 04
Data leakage	Contamination of model inputs with future information	Ch 05

Decision tree	A model that makes predictions via a sequence of yes/no questions (splits) on features	Ch 07
Density forecast	The full probability density function of the predicted variable	Ch 08
DHI	Diffuse Horizontal Irradiance — scattered component of solar radiation	Ch 04
Diebold-Mariano test	A statistical test for comparing forecasting model accuracy	Ch 10
Dispatch price	5-minute price set by the marginal generator	Ch 01
Dispatchable	Generation that can be controlled by the operator (coal, gas, hydro)	Ch 04
Diurnal cycle	Daily repeating pattern in prices	Ch 02
DNI	Direct Normal Irradiance — beam component of solar radiation	Ch 04
Dropout	Regularisation that randomly deactivates neurons during training	Ch 07
Duck curve	Daily net load shape: midday dip, evening spike	Ch 02

E

Early stopping	Stopping training when validation performance plateaus; prevents overfitting	Ch 07
ECMWF	European Centre for Medium-Range Weather Forecasts	Ch 04
Elastic Net	Combination of L1 and L2 penalties	Ch 06

Embargo	Gap between training data and forecast to prevent leakage	Ch 05
Ensemble method	A technique combining multiple weak models into a single stronger model	Ch 07
Epoch	One complete pass through the entire training dataset	Ch 07
ERA5	ECMWF's fifth-generation global reanalysis dataset	Ch 04
Exchangeability	The assumption that data ordering does not affect the joint distribution	Ch 08
Expanding window	Training window that grows over time	Ch 05
Expected value	The probability-weighted average of outcomes across all scenarios	Ch 09
Extrapolation	Making predictions for inputs outside the range observed in training	Ch 10

F

Fair comparison	Evaluating models under identical conditions to ensure performance differences reflect model quality	Ch 10
FCAS	Frequency Control Ancillary Services — grid stability services	Ch 09
FCAS co-optimisation	Jointly optimising energy arbitrage and ancillary service provision	Ch 10
Feature (predictor)	Input variable used by a forecasting model	Ch 05
Feature entry order	Order in which features gain non-zero coefficients as λ decreases	Ch 06

Feature importance	A measure of how much each feature contributes to a tree model's predictions	Ch 07
Feature selection	Identifying which features are useful; automated by LASSO	Ch 06
Forecast combination principle	The finding that averaging forecasts from different models almost always improves accuracy	Ch 08
Forecast horizon	Time span covered by the forecast (48 half-hours = 24 hours)	Ch 05
Forecast skill	Performance relative to a naive baseline; controls for problem difficulty	Ch 10
Forward fill	Imputation method that carries the last known value forward through gaps	Ch 01

G

Gate closure	Deadline by which a forecast must be issued (noon in our setup)	Ch 05
GHI	Global Horizontal Irradiance — total solar radiation on a horizontal surface	Ch 04
Gradient boosting	An ensemble method that sequentially builds trees, each correcting the errors of the previous ones	Ch 07
Grey-box model	A model combining physical structure with data-driven learning	Ch 10
Gross pool	Market design where all electricity must be traded through the central exchange	Ch 01

H

HAC standard error	Standard error corrected for autocorrelation and heteroscedasticity	Ch 05

HDD / CDD	Heating / Cooling Degree Days — thermal stress metrics	Ch 04
Heavy tails	Extreme values occur far more often than a normal distribution predicts	Ch 02
Histogram binning	Discretising continuous features into buckets to speed up split-finding in trees	Ch 07
Hockey stick	Nonlinear net-load-to-price relationship	Ch 03
Hyperparameter	A setting chosen before training (number of trees, learning rate) vs. parameters learned from data	Ch 07

I

Intercept	Constant term in a linear model	Ch 06
Interconnector	Transmission line connecting two NEM regions	Ch 01
Interval-ending	Timestamp convention where rows are labelled with the period's END time	Ch 01
Intra-day re-optimisation	Updating the dispatch schedule multiple times within a day	Ch 10
Isotonic regression	A non-parametric regression constrained to produce monotonically non-decreasing predictions	Ch 10

K

Kurtosis	Measure of tail heaviness; electricity prices have very high kurtosis	Ch 02

L

L1 norm	Sum of absolute values of coefficients	Ch 06
LASSO (L1)	Regularisation with absolute-coefficient penalty; drives some coefficients to zero	Ch 06
LassoCV	LASSO with automatic λ selection via cross-validation	Ch 06
Law of large numbers	Average of many random samples converges to the true expected value	Ch 09
LCOS (Levelised Cost of Storage)	The per-MWh cost of storage over the battery's lifetime	Ch 10
Leaf (terminal node)	The endpoint of a decision tree branch where a prediction is made	Ch 07
LEAR	LASSO Estimated AutoRegressive — per-hour LASSO models for price forecasting	Ch 06
Learning curve	A plot of performance vs. training set size; reveals data sufficiency and overfitting	Ch 07
Learning rate (η)	A small multiplier controlling how much each new tree contributes to the ensemble	Ch 07
LightGBM	A fast, accurate gradient boosting implementation by Microsoft; the industry standard for tabular data	Ch 07
Linear model	A model that predicts output as a weighted sum of inputs — can only represent straight-line relationships	Ch 07
Linear program (LP)	Optimisation with linear objective and constraints	Ch 05
Linear regression	Model predicting target as weighted sum of features	Ch 06
Linke turbidity	Coefficient quantifying atmospheric clarity	Ch 04

Loss differential	Difference in forecast error between two models at each time step	Ch 06

M

MAE	Mean Absolute Error — average of absolute forecast errors	Ch 05
Marginal contribution	The change in performance when a component is added or removed	Ch 10
Marginal cost	Cost of producing one additional unit of output	Ch 03
Marginal pricing	All dispatched generators receive the price of the most expensive unit needed	Ch 01
Marginal value	The additional value from one additional unit of improvement	Ch 10
Market power	Ability of one participant to influence the price	Ch 03
Mean reversion	Tendency of extreme prices to return to a long-run average	Ch 02
Merit order	Ranking of generators from cheapest to most expensive	Ch 03
Merit-order curve	The empirical relationship between net load and electricity price	Ch 10
Merit-order effect	Price depression caused by zero-cost renewables	Ch 03
MLP	Multilayer Perceptron — the simplest fully connected neural network	Ch 07
MPC	Model Predictive Control — rolling re-optimisation strategy	Ch 05

MPC (Model Predictive Control)	A control strategy that repeatedly solves an optimisation over a rolling horizon	Ch 09
MWh	Megawatt-hour — standard unit of energy in wholesale markets	Ch 01

N

NEM	National Electricity Market — Australia's wholesale electricity market for the eastern states	Ch 01
Net load	Demand minus variable renewable generation	Ch 05
Non-stationarity	The property of a time series whose statistics change over time	Ch 07
Nonlinear model	A model that can learn curved, threshold, or discontinuous relationships	Ch 07
NWP	Numerical Weather Prediction — physics-based weather forecasting	Ch 04

O

OCGT	Open Cycle Gas Turbine — fast-start but expensive peaker	Ch 03
OLS	Ordinary Least Squares — standard coefficient estimation method	Ch 06
Open-loop control	Executing a fixed plan without adjustment	Ch 05
Overconfidence	When prediction intervals are too narrow (stated coverage exceeds actual coverage)	Ch 08
Overfitting	When a model memorises training data specifics instead of learning generalisable patterns	Ch 07

P

Parquet	Columnar data storage format; standard for analytical workloads	Ch 01
PDF (probability density function)	A function showing the relative likelihood of each possible value	Ch 08
Per-hour models	Separate model for each hour of the day	Ch 06
Perfect foresight	Theoretical optimum with known future prices	Ch 05
Perfect foresight revenue	Maximum arbitrage revenue achievable with perfect knowledge of future prices	Ch 09
Persistence forecast	Predicting tomorrow = today	Ch 04
Pinball loss	The asymmetric loss function used to train quantile regression models	Ch 07
Pipeline	A sequence of processing stages where each stage's output becomes the next stage's input	Ch 10
Point forecast	A single-valued prediction with no uncertainty information	Ch 07
Pre-dispatch	AEMO's projection of future dispatch and prices	Ch 05
Prediction interval	A range expected to contain the actual value with a specified probability	Ch 08
Price spike	Abrupt, extreme, temporary price increase	Ch 03
Price-duration curve	Prices sorted from high to low vs % of time exceeded	Ch 02

Price-taker assumption	The assumption that a participant's actions do not affect market prices	Ch 10
Probabilistic forecast	A forecast that describes the full range of possible outcomes and their probabilities	Ch 08
pvlib	Python library for solar energy system simulation	Ch 04

Q

QRA (Quantile Regression Averaging)	A method combining multiple point forecasts into calibrated quantile forecasts	Ch 08
Quantile crossing	When independently predicted quantiles violate their natural ordering	Ch 07
Quantile forecast	Predictions at multiple probability levels characterising the forecast distribution	Ch 08

R

Rated capacity	Maximum sustained turbine output	Ch 04
Reanalysis	Physics-model reconstruction of historical atmosphere using all available observations	Ch 04
Regime change (structural break)	A fundamental change in the data's statistical properties	Ch 10
Regularisation	Any technique that prevents a model from fitting noise in the training data	Ch 07
Regularisation path	Sequence of LASSO solutions as λ varies	Ch 06
Reliability diagram	A plot of observed coverage vs. nominal coverage for assessing calibration	Ch 08

Residual	The difference between the actual value and the current prediction; what the next tree tries to correct	Ch 07
Revenue stack	The combination of all revenue streams available to a battery	Ch 10
Ridge (L2)	Regularisation with squared-coefficient penalty; shrinks but keeps all features	Ch 06
rMAE	Relative MAE — model MAE divided by naive MAE	Ch 05
RMSE	Root Mean Square Error — penalises large errors more	Ch 05
RNN / LSTM / GRU	Recurrent architectures that process sequential data via internal memory	Ch 07
Robust optimisation	Optimisation that protects against worst-case outcomes	Ch 09
Rolling recalibration	Periodically recomputing conformal corrections as new data arrives	Ch 08
Rolling-origin backtest	Walk-forward evaluation across multiple train/test splits	Ch 05
Rolling-origin evaluation	Repeatedly training and testing across multiple time windows	Ch 10
Round-trip efficiency	Fraction of stored energy recoverable on discharge	Ch 05
Round-trip efficiency (RTE)	Fraction of energy recovered from a charge-discharge cycle	Ch 09

S

Sample efficiency	How much data a model needs to reach its potential performance	Ch 07

Scenario	One complete realisation of uncertain future prices	Ch 09
Scenario forecast	Multiple complete future trajectories sampled from the forecast distribution	Ch 08
Seasonality	Regular patterns repeating at fixed intervals	Ch 02
SHAP values	Game-theory-based feature attribution that explains individual predictions	Ch 07
Sharpness	How narrow (concentrated) the prediction intervals are	Ch 08
Similar-day naive	Forecast using same half-hour, same day of week, one week prior	Ch 05
Sliding window	Training window of fixed length that moves forward	Ch 05
SOC	State of charge — current stored energy in the battery	Ch 05
Solar zenith angle	Angle between the sun and directly overhead	Ch 04
Sparsity	Property of having most coefficients exactly zero	Ch 06
Spike coverage	Fraction of extreme price events captured within the prediction interval	Ch 08
Split conformal prediction	A variant using a held-out calibration set to compute interval corrections	Ch 08
Split point	A feature-value threshold that divides data into two groups at a tree node	Ch 07
Stationarity	Statistical properties constant over time	Ch 06

Stochastic programming	Optimisation under uncertainty using multiple scenarios	Ch 09
Strategic bidding	Setting bids based on market conditions rather than cost	Ch 03
Stylised facts	Consistent empirical patterns in a dataset	Ch 02
Supply stack	Visual representation of cumulative generation vs bid price	Ch 03
Synoptic scale	Weather features spanning hundreds to thousands of km	Ch 04

T

Temporal arbitrage	Exploiting price differences across time via storage	Ch 05
Time-series CV	Cross-validation respecting temporal ordering	Ch 06
Trading day	NEM day from 04:00 to 04:00 AEST	Ch 05
Trading price	30-minute average of six dispatch prices; the financially settled price	Ch 01
Transformer	An architecture using self-attention to capture long-range dependencies in parallel	Ch 07
Two-stage stochastic program	Decisions made before uncertainty resolves (first stage); outcomes computed after (second stage)	Ch 09

U

Underconfidence	When prediction intervals are too wide (actual coverage exceeds stated coverage)	Ch 08

V

Value of information	The economic benefit of having a better forecast	Ch 09
Variance	Error from model sensitivity to specific training data	Ch 06
Volatility	Degree of price variation over time	Ch 02
Volatility clustering	Tendency for volatile periods to persist	Ch 02
VRE	Variable Renewable Energy — generation whose output depends on weather (wind, solar)	Ch 04

W

White-box model	A model built from explicit physical or economic principles	Ch 10
Wind shear	Increase in wind speed with altitude	Ch 04